

Initial Investigation into Regime Detection Neural Network Signal for Trading Strategy in Equity, FX and Rates Futures

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Executive Summary

A neural network (NN) has been trained on market features to detect daily underlying market regime and consequently offered to a trading firm as a tool to complement daily trade conviction. Individual NNs are available for several asset classes, targeting 3-month futures trading available via the Chicago Mercantile Exchange (CME). The provided signal is a binary signal declaring bullish or bearish expected trends, and is issued pre-market opening each trading calendar day.

An initial investigation was conducted into whether the signal could be utilised to underpin a trading strategy, rather than just complement other available conviction signals. The strategy assumed that trade will take a daily position in the pre-opening auction markets and close either at closing auction or upon regime reversal being detected. The investigation was interested in the signal's hit rate, average daily returns, cumulative returns, strategy volatility, and Sharpe ratio, as initial metrics of the strategy's potential for further development. Liquidity was assumed available and transaction costs neglected.

The analysis demonstrated that the signal has inconsistent and poor accuracy across the asset classes investigated. While some periods of trading were profitable, the inconsistency of the returns poses significant risk and is discouraging of the strategy's further development. Should development continue, future investigations should evaluate whether the signal holds potential in the opening period of trade (prior to erosion of the alpha signal) and/or can be improved by hedging of the downside risk.

Quantitative Skills/Knowledge Demonstrated

This project conducts a standard strategy performance analysis, through back-testing of a signal applied to historical market pricing data. It demonstrates command of Python (and associated packages) for enacting trading logic, sourcing historical data (yfinance), large dataset cleaning and exploration (Pandas), analytics (NumPy and Statistics), and data communication/plotting (Matplotlib) in a Microsoft Visual Studio and OS environment. The performance of the strategy is consequently reported in standard commercial metrics such as hit rates, daily log returns, cumulative returns, trade counts, volatility, drawdown, Sharpe ratio and Calman ratios.

Introduction

A neural network (NN) has been trained on market features to detect daily underlying market regime in an attempt to provide trading edge and conviction to its market participant owners. Several separately trained NNs are available for a range of equity index, FX and rates futures. The purpose of this investigation is to evaluate an initial potential for the daily pre-market signal to be used as a principle strategy signal, rather than a complementary one (as initially intended by its developers).

For the evaluation, the developers have provided historical data for several of the available assets:

1. E-mini S&P 500 3-month Future (ES)
2. E-mini Nasdaq-100 3-month Future (NQ)
3. U.S. 10-year Treasury Note (ZN)
4. Euro-Bobl (FGBM)
5. Euro FX (6E)

The length of available historical data varies with each asset due to their commission and activation dates. Data provided was timestamped for authenticity and auditability. The provided signal is binary (bullish or bearish) and thus it is assumed daily trading will consequently take long and short positions respectively. It is assumed that trades are open/closed at the historical open/close market signal - these are equivalent to trading every market morning without price slippage and/or overnight movement. Brokerage costs are ignored, as is capital/volume evaluation.

The initial investigation aims only to establish whether the signal has further potential for strategy development, or likely lacks the necessary promising metrics. The analysis will therefore compile, clean and format available historical signal and market data, calculate daily trade performance and subsequently report its findings.

Method

The following investigative steps will be taken to address the aforementioned questions:

1. Compile separate historical signal data into single coherent data frame, addressing formatting, errors and indexing.
2. Source and compile market historical data to reconcile with historical signal data, standardising formatting and indexing.
3. Merge historical signal and market data into single Dataframe, allowing for subsequent analysis of each trading asset.
4. For each asset, calculate historical performance and report calculated figures through suitable visual plots / tables.

Analysis

1) E-mini S&P 500 3-month Future (ES)

The historical signal data for asset marker "ES" was the most extensive provided, with an auditable record since its inception in 2020. As a model, it has undergone several period of development across the past few years and is now approximately 1.5 years into it's latest development cycle. Presently, the model produces daily signals on the present 3-month futures contracts until a final 5-day period prior to settlement. It is trained in-between daily signal production and using the 5-day period for planned institutional and infrastructural maintenance.

Figure 1 illustrates the strategy's performance against a long market position across a corresponding period. Flat periods in the strategy's returns are non-trading windows where the model is undergoing further development and training. A total of 782 trading days were observed, with a daily signal hit rate of 52.17%. The strategy experienced a 19.21% maximum drawdown, annualized return of 11.65%, Sharpe of 1.08 and Calmar of 0.606. Its performance by trade quarter is summarised in Figure 2.

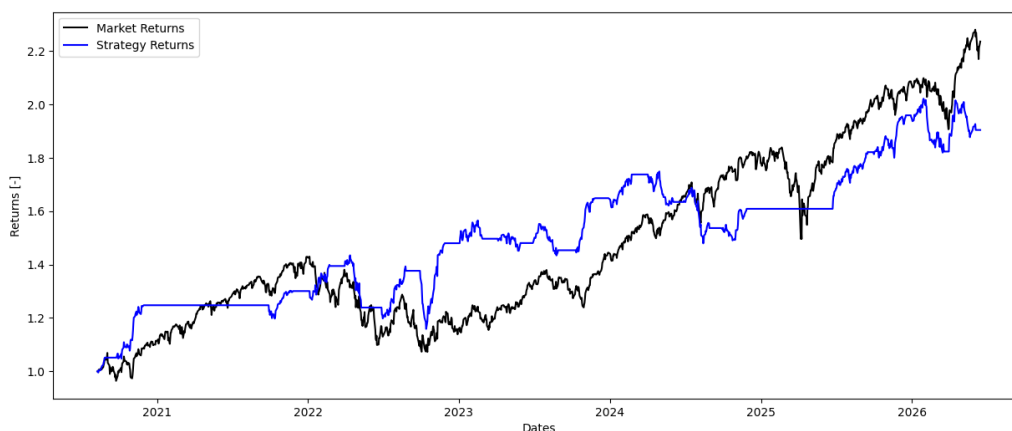


Figure 1: Backtest of ES trading strategy using NN signal for entry/exit

Quarter	Hit_Rate	Avg_Return	Total_Return	Volatility	Trades	Sharpe
2020Q2	53.33%	0.0012	0.0550	0.0088	45	1.89
2020Q3	76.92%	0.0034	0.0478	0.0040	14	12.95
2020Q4	55.56%	0.0037	0.1673	0.0108	45	5.21
2021Q4	57.14%	0.0009	0.0386	0.0069	42	1.71
2022Q1	54.55%	0.0022	0.0728	0.0119	33	2.69
2022Q2	42.86%	-0.0042	-0.1189	0.0158	28	-4.46
2022Q3	52.38%	0.0028	0.1172	0.0107	42	3.89
2022Q4	51.22%	0.0017	0.0677	0.0178	41	1.31
2023Q1	56.76%	0.0006	0.0210	0.0097	37	0.63
2023Q2	29.73%	-0.0008	-0.0305	0.0072	37	-2.22
2023Q3	45.24%	-0.0006	-0.0242	0.0070	42	-1.72
2023Q4	66.67%	0.0038	0.1249	0.0077	33	7.39
2024Q1	51.28%	0.0013	0.0523	0.0069	39	2.67
2024Q2	46.51%	-0.0012	-0.0518	0.0078	43	-2.80
2024Q3	51.11%	-0.0011	-0.0504	0.0100	45	-2.06
2024Q4	53.49%	0.0010	0.0411	0.0071	43	1.75
2025Q3	60.66%	0.0020	0.1244	0.0066	61	4.51
2025Q4	57.14%	0.0012	0.0687	0.0077	56	2.16
2026Q1	42.86%	-0.0013	-0.0719	0.0090	56	-2.57

Figure 2: Quarterly summarised trading statistics for marker "ES"

2) E-mini Nasdaq-100 3-month Future (NQ)

The historical signal data for asset marker "NQ" spanned only the two quarters of 2026. The NN model is subject to the same development and delivery approach as marker "ES". Back-tested quarterly results are summarised in Figure 3. The limited historical data restricts the ability to observe any persistent performance from the signal. Its performance in Q2 2026 is promising.

Quarter	Hit_Rate	Avg_Return	Total_Return	Volatility	Trades	Sharpe
2026Q1	48.28%	-0.0008	-0.0484	0.0117	58	-1.38
2026Q2	66.67%	0.0051	0.2337	0.0105	48	7.40

Figure 3: Quarterly summarised trading statistics for marker "NQ"

3) U.S. 10-year Treasury Note (ZN)

The historical signal data for asset marker "ZN" spans three quarters of trading. The NN model is subject to the same development and delivery approach as marker "ES". Back-tested quarterly results are summarised in Figure 4. The model's daily accuracy averages at slightly over 50% and results in an underwhelming Sharpe result.

Quarter	Hit_Rate	Avg_Return	Total_Return	Volatility	Trades	Sharpe
2025Q4	52.63%	0.0002	0.0139	0.0023	57	0.45
2026Q1	48.28%	0.0002	0.0122	0.0027	58	0.21
2026Q2	54.17%	0.0004	0.0194	0.0033	48	1.17

Figure 4: Quarterly summarised trading statistics for marker "ZN"

4) Euro-Bobl (FGBM)

With only a single trading quarter of historical data provided, the asset marker "FGBM" is of little consequence to the overall study. Furthermore, it's signal accuracy and overall metrics were poor. Back-tested results are summarised in Figure 5. The NN model is subject to the same development and delivery approach as marker "ES".

Quarter	Hit_Rate	Avg_Return	Total_Return	Volatility	Trades	Sharpe
2025Q3	47.17%	-0.0002	-0.0083	0.0015	53	-3.64

Figure 5: Quarterly summarised trading statistics for marker "FGBM"

5) Euro FX (6E)

The historical signal data for asset marker "6E" covers almost a year of trading. The NN model is subject to the same development and delivery approach as marker "ES". Back-tested results are summarised in Figure 6. The signal results in a poor hit rate, however the poor performance is more consistent than other markers. An analysis was consequently undertaken to reverse the signal in the hope of improved accuracy and performance. Performance gains from the reversed signal were marginal, highlighting an imbalance in the signal's accuracy in predicting bullish and bearish trends.

Quarter	Hit_Rate	Avg_Return	Total_Return	Volatility	Trades	Sharpe
2025Q3	43.10%	0.0001	0.0080	0.0054	58	-0.11
2025Q4	43.40%	-0.0004	-0.0200	0.0027	53	-3.31
2026Q1	37.74%	-0.0007	-0.0374	0.0045	53	-3.13
2026Q2	48.84%	0.0003	0.0104	0.0031	43	0.37

Figure 6: Quarterly summarised trading statistics for marker "6E"

Summary of Findings

The initial investigation results highlight inconsistent returns and hit rates when using the provided signals as strategy leads under back-testing. These results suggest that the raw signals would require substantial further development and constraining to potentially result in a strategy with consistently positive returns and reduced volatility. The results therefore indicate that further investigation is likely not warranted, due to a decreased likelihood of sufficient returns being realised under real market conditions (slippage, transaction costs, etc.). It is possible that the signals are sufficient under different trading parameters (e.g., intra-day).

Further Investigations

The following further investigative steps are recommended:

1. It is noted via prior experience that pre- market opening signals typically see a decay in their forecast capacity throughout the day - their purported 'alpha' can decay rapidly within the opening minutes/hours of trade. The adopted multi-day / whole-day approach may have suffered from these affects. An investigation into the potential of a successful trading strategy from the opening minutes/hours of trade may be warranted.
2. Inter-asset relationships (i.e. correlation) were not explored due to a limit in historical signal data restriction the significance of any findings. It is possible that paired trading or even paired-signals could be utilise to improve the prospects of returns.